**Suggested problems:** 1-5, 6(a), 7, 8

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**Problem 1:** Let a coin be flipped two times with a probability $p = 0.75$ of obtaining heads. Let $X$ be the number of heads obtained on the first flip, and $Y$ the total number of heads obtained on both flips. Describe the joint probability distribution of $X$ and $Y$.

**Problem 2:** A fair coin is flipped three times. One of the variables of interest is $Y$, the total number of heads obtained on the three flips. Let $X$ denote the amount of money won on a side bet in the following manner. If the first head occurs on the first toss, you win \$1. If the first head occurs on flip 2 or flip 3 you win \$2 or \$3, respectively. If no head appears, you lose \$1 (that is, win $-$\$1.)

(a) Find the joint probability function for $X$ and $Y$.

(b) What is the probability that you will win \$1 or less and fewer than three heads will occur?

**Problem 3:** (Example 3.4.7) Suppose that $X$ and $Y$ are jointly continuous with density function

$$f(x,y) = \begin{cases} cx^2y & : x^2 \leq y \leq 1, \\ 0 & : \text{otherwise}, \end{cases}$$

for some constant $c$.

(a) Compute $c$.

(b) Compute $P(X \geq Y)$.